YILI YANG, PhD

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SKILLS

DOMAINS

Geospatial Information Science
Remote Sensing|Geosciences
Digital Rocks|AI for Science
Scientific Data & Image Processing
Machine Learning & Deep Learning
Multimodal Learning
Image Segmentation|Object Detection
Time-series modelling
Earth Foundation Models

TECH STACK

Google Earth Engine|QGIS
Google Cloud Platform|ArcGIS Pro
Python|PyTorch|Scikit-learn
NumPy|pandas|SciPy|TensorFlow

ENGINEERING

Agentic Coding|Docker|GitHub Actions
Harness Engineering|Claude Code/skills
Prompt and Context Engineering
Git|GitHub|Google Colab|VScode

EDUCATION

UNIVERSITY OF EDINBURGH

PhD Petrophysics,2021
MSc Geology, 2016
BSc Geoscience, 2015

STATS

PUBLICATIONS

Journal Papers: 10
Conferences: 9
Data Sets: 2

ACADEMIC SERVICES

Invited Talks & Panels: 17
Peer-reviews: 8
Mentees: 9

LINKS

Personal Website
LinkedIn || ORCID
Google Scholar || GitHub

EXPERIENCE

AI SCIENTIST -- GEOSPATIAL & REMOTE SENSING | Woodwell Climate Research Center

Jan 2022 - Present | Falmouth, MA, USA

- **Project Lead**: Thaw slumps detection using deep learning and remote sensing
- **Project Lead**: Circumpolar thaw slump probability map, released open at the Arctic Data Center
- **Project Lead**: Multi-domain climate time-series prediction
- **Collaboration**: Computer vision wild fire detection
- **Collaboration**: Machine learning wild fire prediction
- **Collaboration**: Carbon Flux time-series data imputation
- **Collaboration**: Prediction of thaw slump initialisation
- **Collaboration**: Satellite image segmentation with Location Embedding
- **Collaboration**: Permafrost thaw and Carbon Emission
- **Workshops**: Annual Workshop for Machine Learning

DATA SCIENCE FELLOW | Faculty.AI

Sep 2021 - Dec 2021 | London, UK

- Technical and commercial training for professional data scientist followed by work placement on a client project

DOCTORAL RESEARCHER |ICCR

International Centre for Carbonate Reservoirs
Sep 2017 - Sep 2020 | Edinburgh, UK

- Petrophysical calculation and simulation algorithm development
- Deep learning image segmentation for digital rocks
- Synchrotron X-ray μ-CT multiphase fluid experiments

AWARDS & GRANTS

FUNDING FOR CLIMATE SOLUTIONS

Funding for Climate Solutions, Woodwell Climate Research Center
$99,749| 2025-2026 | A Generic Climate AI Framework for Multi-domain Time Series Prediction

MULTIDISCIPLINARY COLLABORATIONS

PETROBRAS × U OF EDINBURGH

Understanding how oil and water flow in rocks

GOOGLE × WOODWELL

Understanding Permafrosts, AlphaEarth and Satellite Embedding

ESRI × WOODWELL

Understanding Permafrosts and Indigenous Communities

MICROSOFT × WOODWELL

Planetary Computer, permafrost data & compute (early stage)

ACADEMIC COLLABORATIONS

The Chinese University of Hong Kong | University of Connecticut
University of Alaska Fairbanks | Arizona State University
University of Illinois | Alfred Wegener Institute | Harvard Kennedy School